Module 3
Uninformed Search

Nirmala Shinde Baloorkar

Assistant Professor

Department of Computer Engineering

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Outline

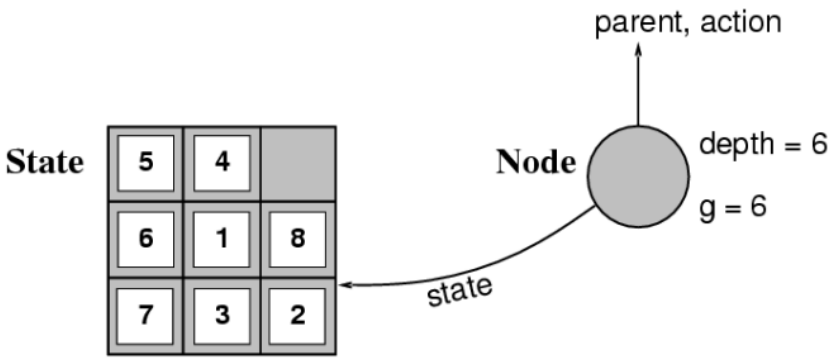
- Blind search strategies
  - Breadth-first search
  - Depth-first search
  - Depth-limited search
  - Iterative deepening search
- For each of these strategies
  - The algorithm
  - The time and space complexities
  - When to select a particular strategy

Uninformed ("Blind") Search Strategies

- No information is available other than
  - The current state
    - Its parent (perhaps complete path from initial state)
    - Its operators (to produce successors)
  - The goal test
  - The current path cost (cost from start state to current state)

Terms

- **Node**: Represents a state in the problem along with metadata (e.g., path cost, parent, action).



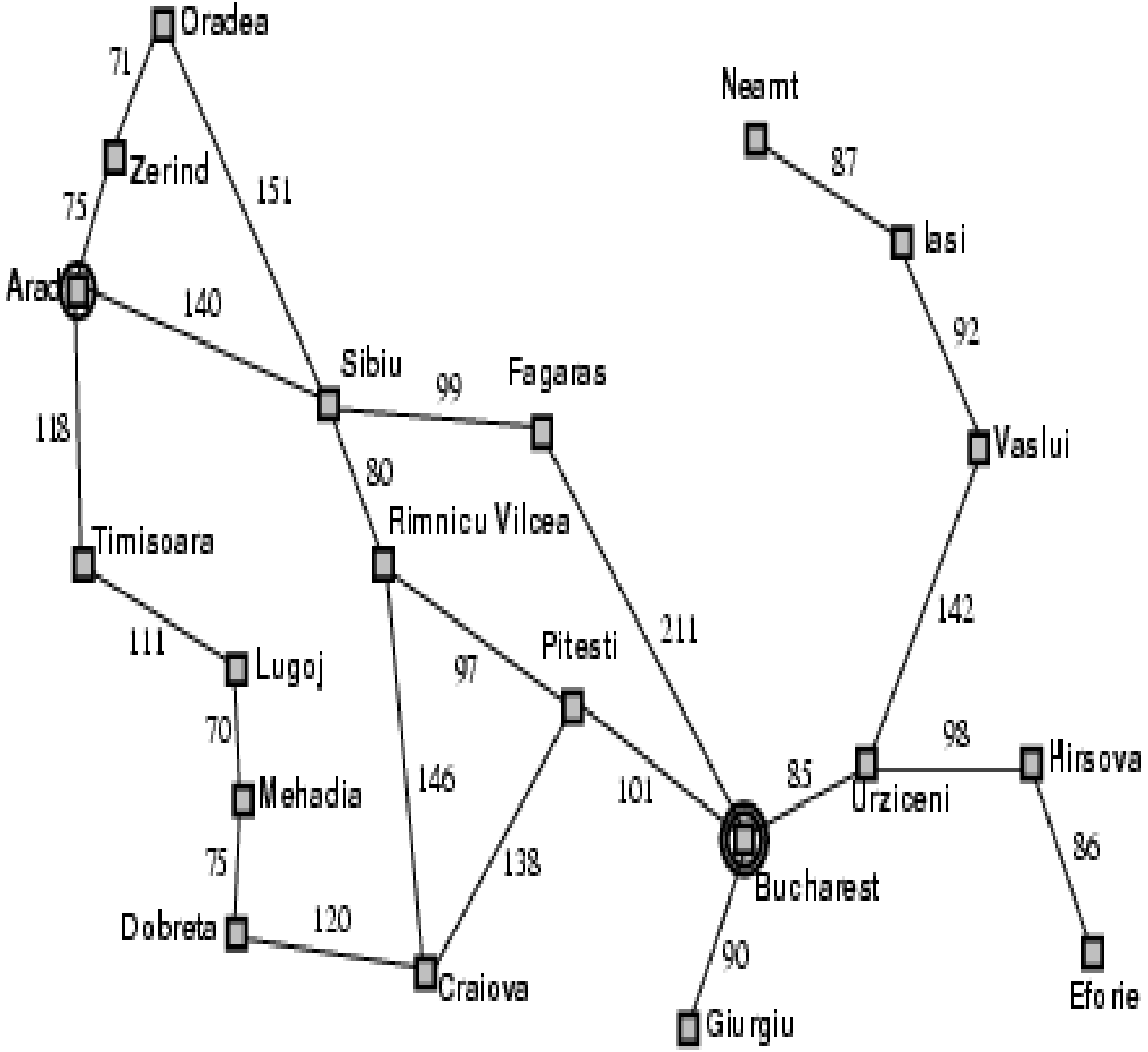
- **Expansion**: Generating successors from a node using valid actions.
- **Fringe**: The set of nodes waiting to be explored.

Handling repeated states

- Initialize the **fringe** using the **starting/initial state**
- While the fringe is not empty
  - Choose a fringe node according to **search strategy** and take it off the fringe
  - If the node contains the **goal state**, return solution
  - Else **expand** the node and add its children to the fringe

- To handle repeated states:
  - Every time you expand a node, add that state to the **explored set**; do not put explored states on the fringe again
  - Every time you add a node to the fringe, check whether it already exists in the fringe with a higher path cost, and if yes, replace that node with the new one

Example: I


Oradea
71
Zerind
151
75
Arad
140
Sibiu
99
Fagaras
118
80
Rimnicu Vilcea
Timisoara
111
Lugoj
97
Pitesti
211
70
Mehadia
146
101
75
138
Dobreta
120
Craiova
Bucharest
90
Giurgiu
Neamt
87
Iasi
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Vaslui
142
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Urziceni
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Hirsova
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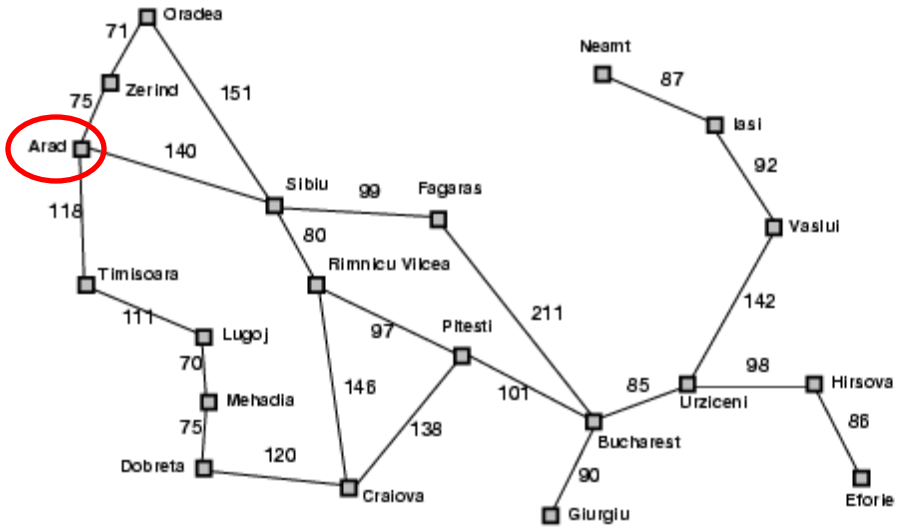
Eforie

Search without repeated states



Start: Arad
Goal: Bucharest

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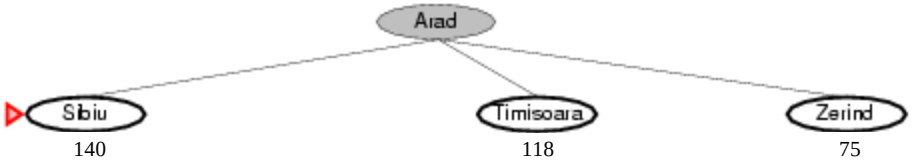
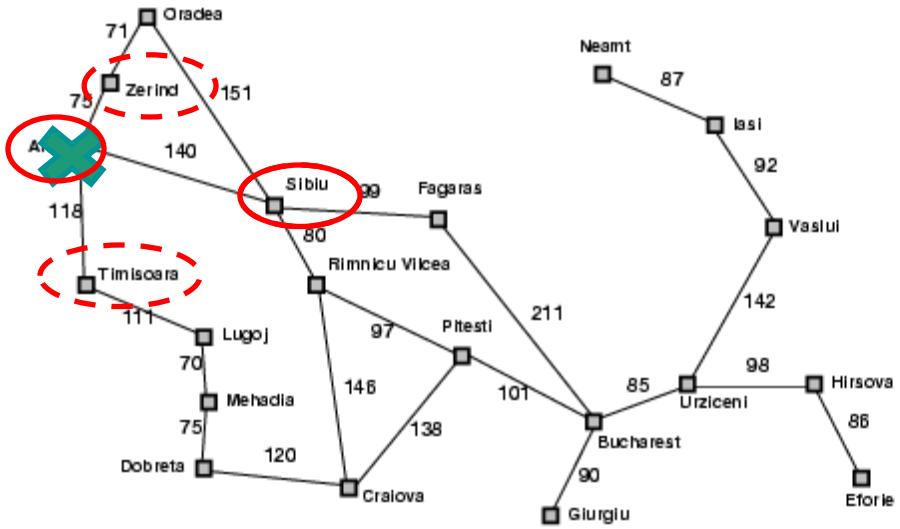
Search without repeated states





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Goal: Bucharest

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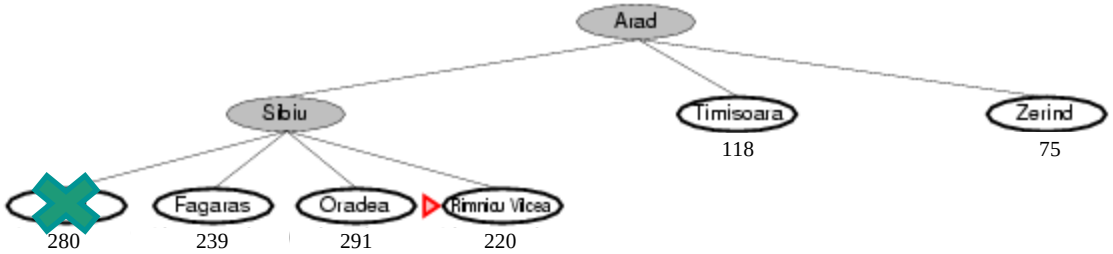
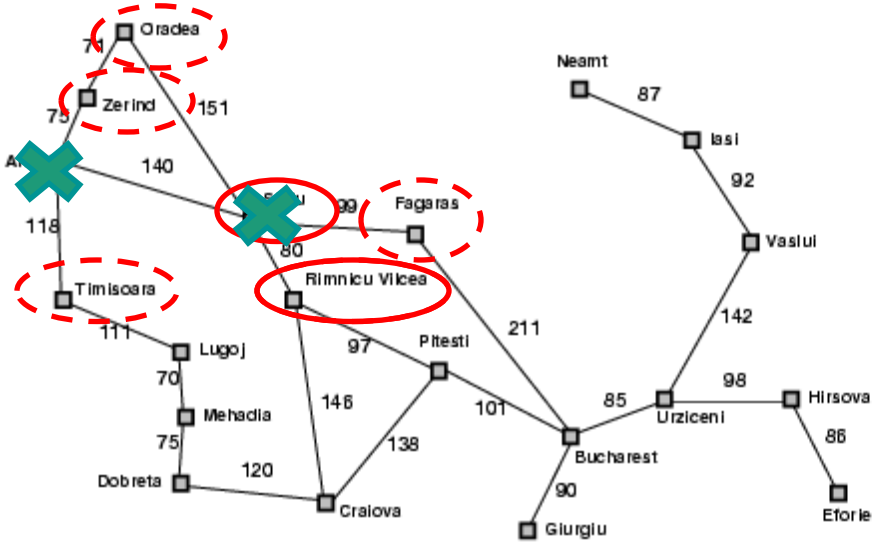
Search without repeated states





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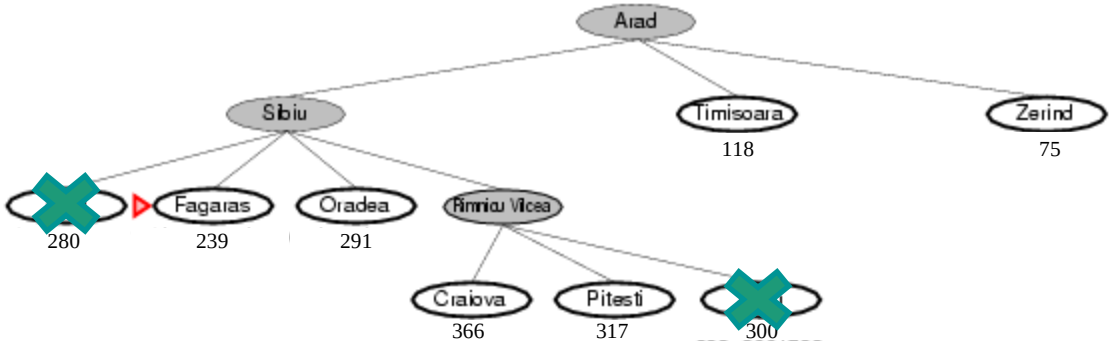
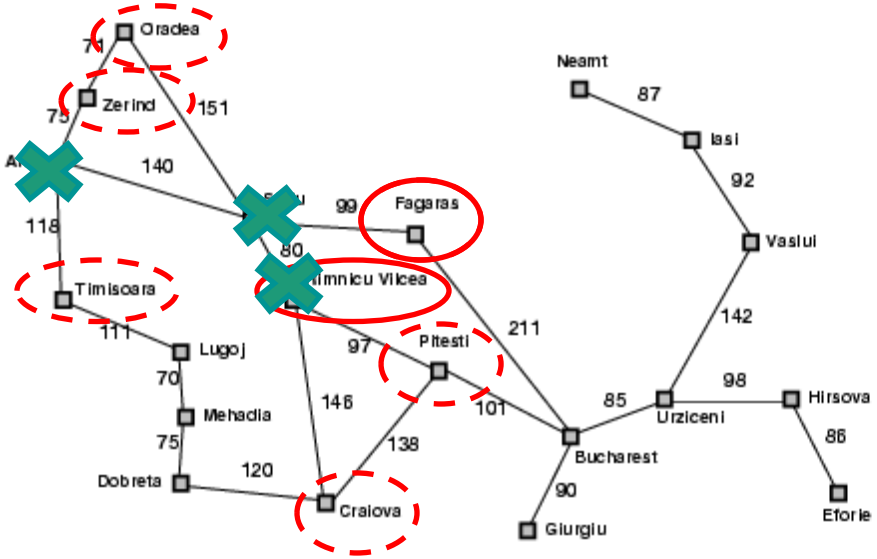
Search without repeated states





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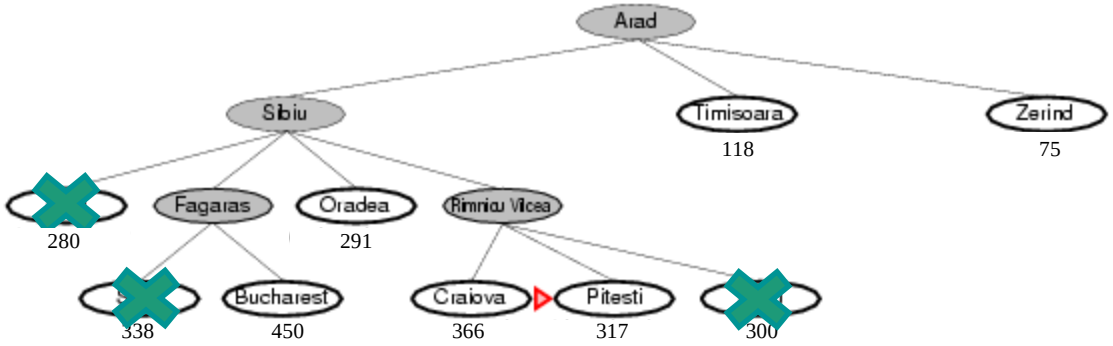
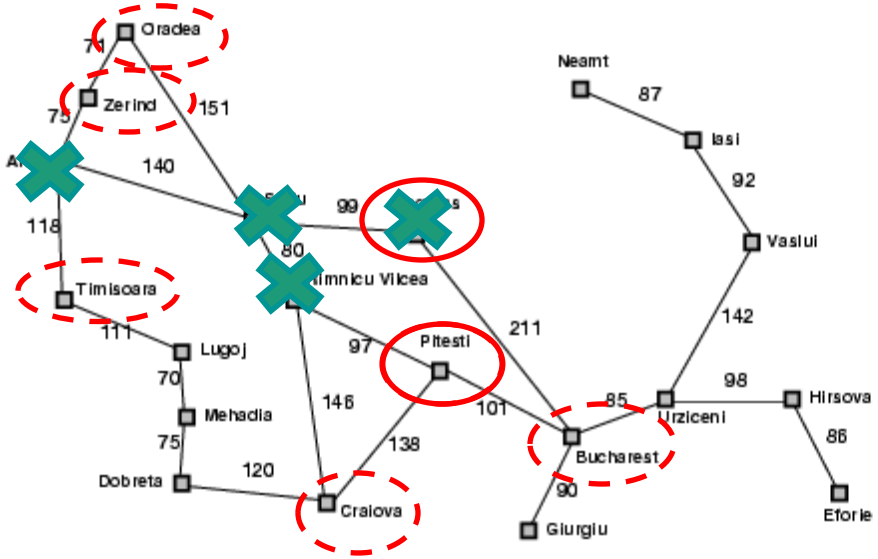
Search without repeated states





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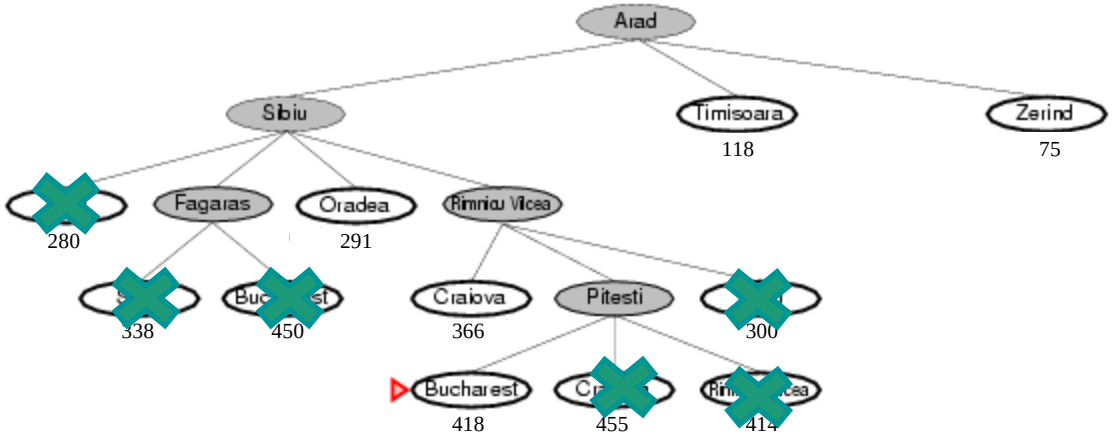
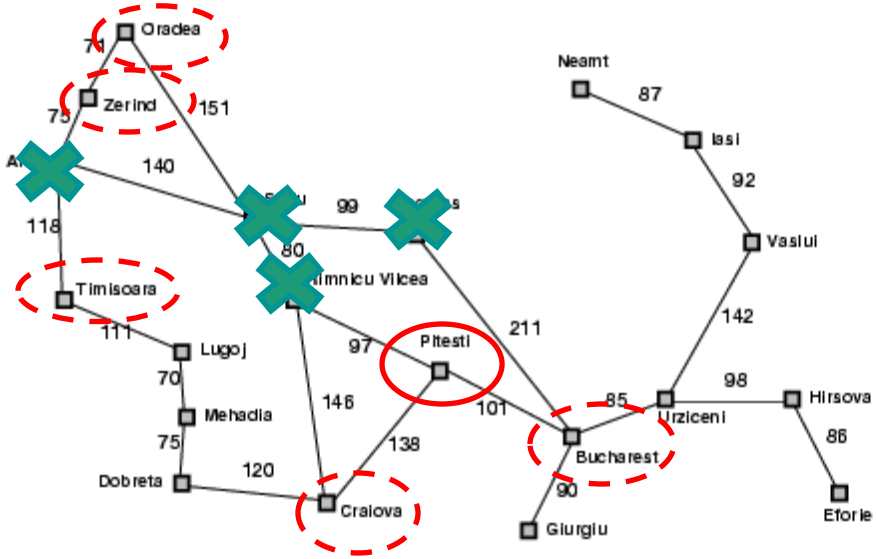
Search without repeated states





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Goal: Bucharest

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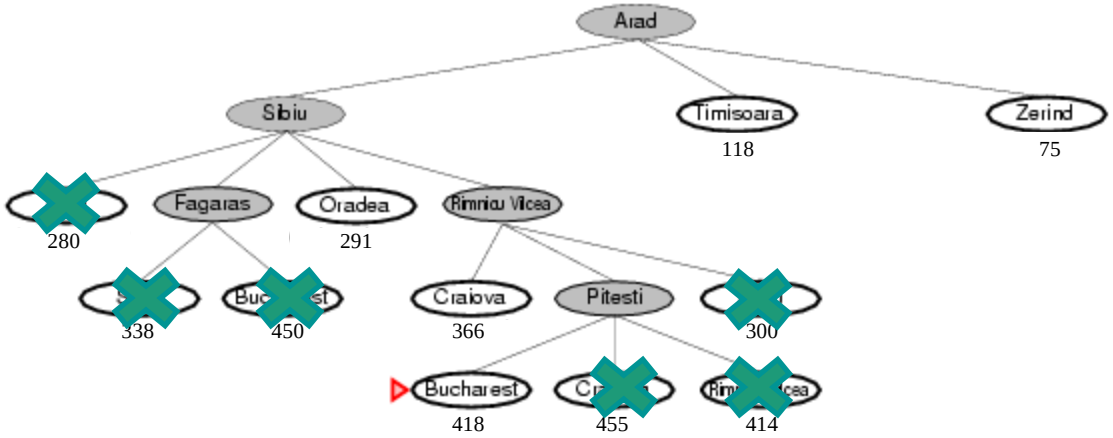
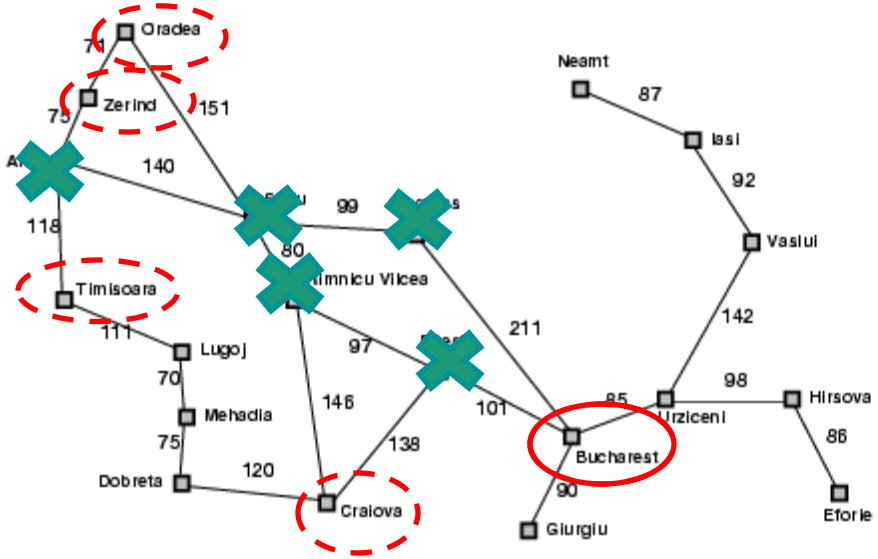
Search without repeated states





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Goal: Bucharest

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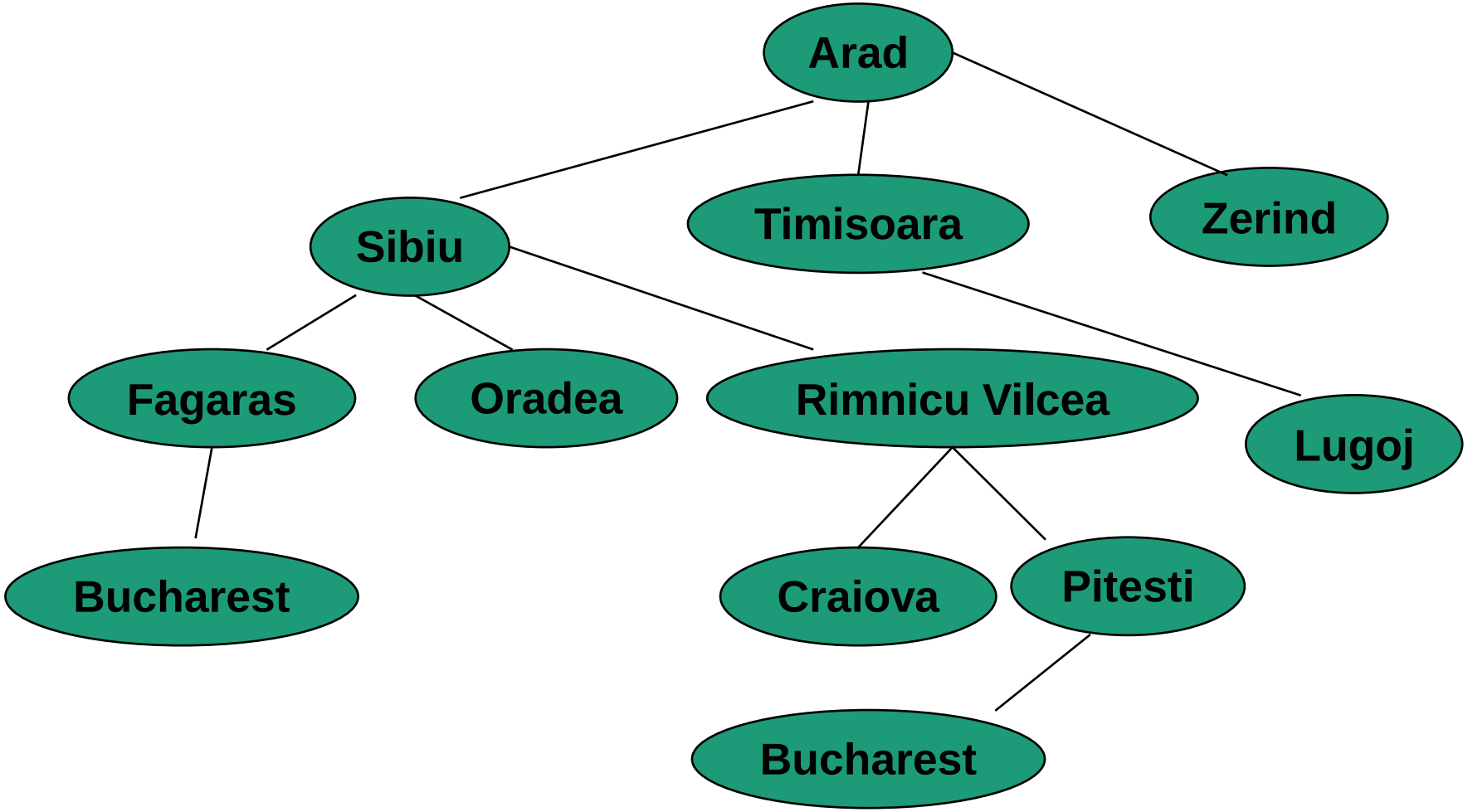
Breadth First Search

Algorithm

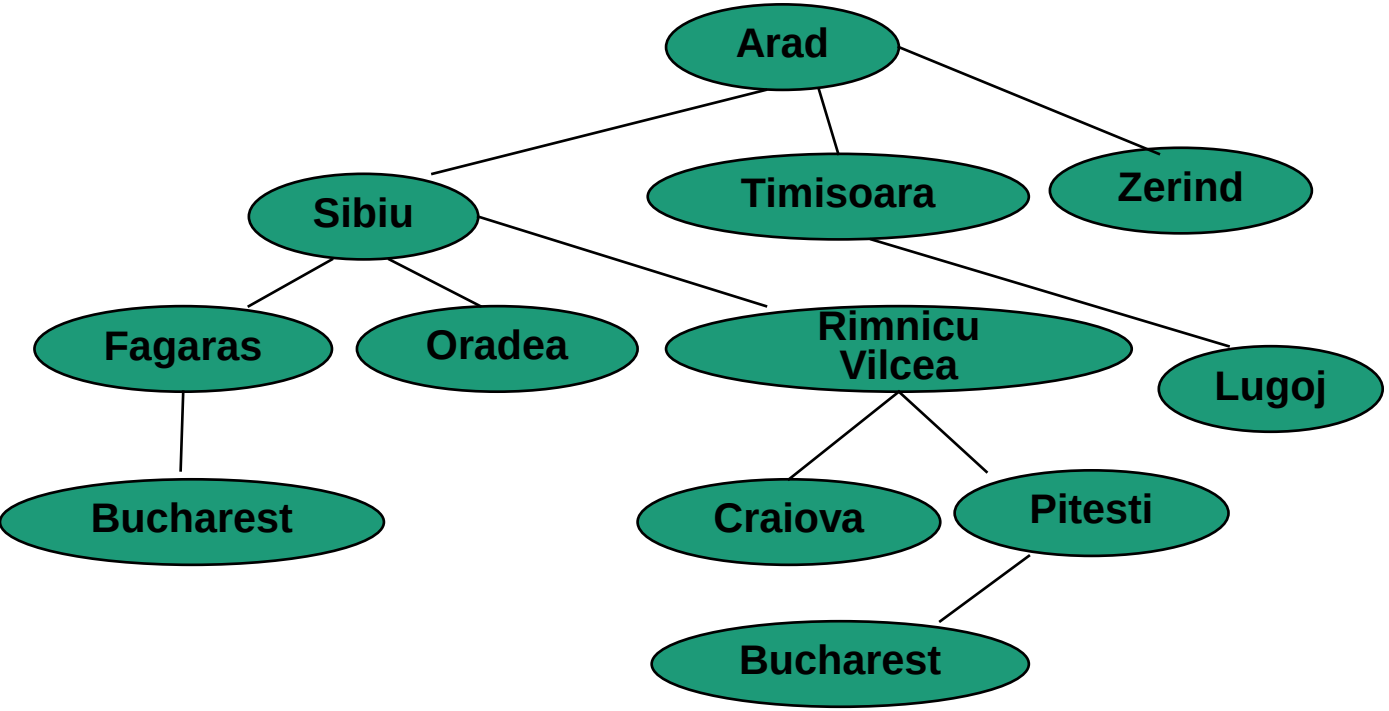
```
Let fringe be a list containing the initial state

loop do
    if fringe is empty then return failure
    Node ←— remove-first(fringe)
    if Node is goal state then
        return the corresponding solution
    else expand all successor of Node and
            add the resulting(newly generated) nodes to the Back of
            fringe
  end
```

Node will be expanded in a FIFO order.

Example – Search Tree



BFS



Fringe	Closed
(A)	Nil
(S T Z)	(A)
(T Z F O RV)	(A S)
(Z F O RV L)	(A S T)
(F O RV L)	(A S T Z)
(O RV L B)	(A S T Z F)
(RV L B)	(A S T Z F O)
(L B C P)	(A S T Z F O RV)
(B C P)	(A S T Z F O RV L)
(C P)	Goal Node Found

Path : Arad-Sibiu-Timisoara-Zerind-Fagaras-Oradea-Rimnicu Vilcea-Lugoj-Bucharest

Breadth-First Search

- Complete? Yes
- Optimal? Yes
- Time complexity? Exponential: $O(b^d)$
- Space complexity? Exponential: $O(b^d)$

In practice, the memory requirements are typically worse than the time requirements

Breadth First Search

- Advantages
  - Finds the path of minimal length to the goal.
- Disadvantages
  - Requires the generation and storage of a tree whose size is exponential the depth of the shallowest goal node

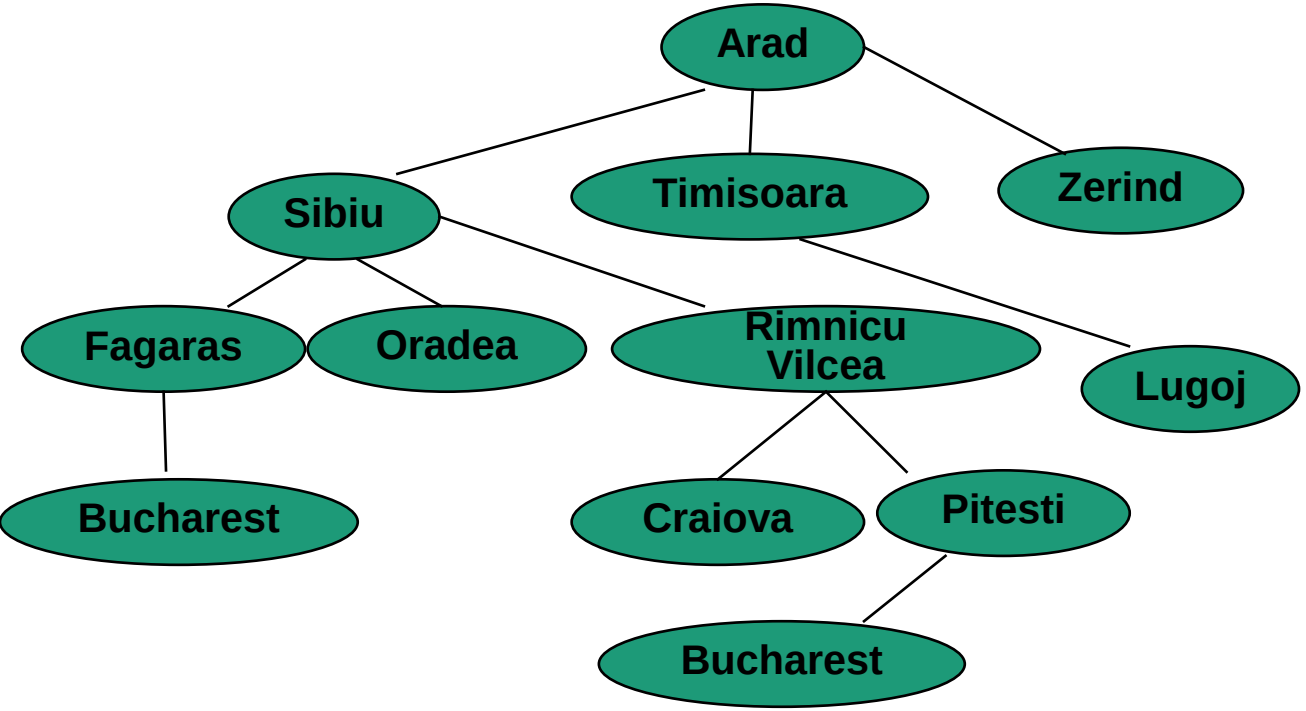
Depth First Search

Algorithm

```
Let fringe be a list containing the initial state

loop do
    if fringe is empty then return failure
    Node ←— remove-first(fringe)
    if Node is goal state then
        return the corresponding solution
    else expand all successor of Node and
            add the resulting(newly generated) nodes to the front of
            fringe
  end
```

Node will be expanded in a LIFO order.

DFS



Fringe	Closed
(A)	Nil
(Z T S)	(A)
(Z T RV O F)	(A S)
(Z T RV O B)	(A S F)
(Z T RV O)	Goal Node Found

Path : Arad-Sibiu-Fagaras-Bucharest

Properties of depth-first search

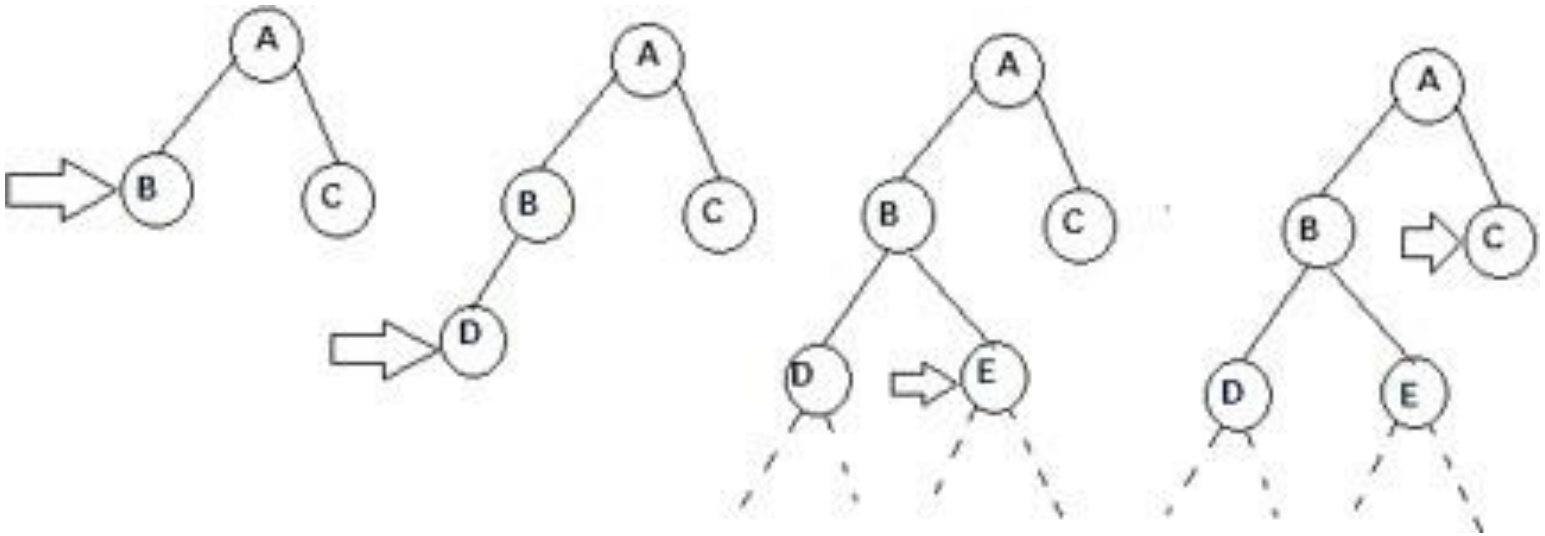
- **Complete?**
  Fails in infinite-depth spaces, spaces with loops
  Modify to avoid repeated states along path
  - complete in finite spaces
- **Optimal?**
  No – returns the first solution it finds
- **Time?**
  Could be the time to reach a solution at maximum depth m: $O(b^m)$
  Terrible if m is much larger than d
  But if there are lots of solutions, may be much faster than BFS
- **Space?**
  $O(bm)$, i.e., linear space!

Depth Limited Search (DLS)

- Depth-limited search is a variant of depth-first search designed to handle infinite state spaces by imposing a predetermined depth limit. In this approach, nodes at the specified depth are treated as if they have no successors.
- **Termination Conditions**
  - A solution is found within the depth limit.
  - No solution exists within the given depth limit.

Depth Limited Search (DLS)

- If depth is fixed to 2, DLS carries out depth first search till second level in the search tree.



Depth Limited Search (DLS)

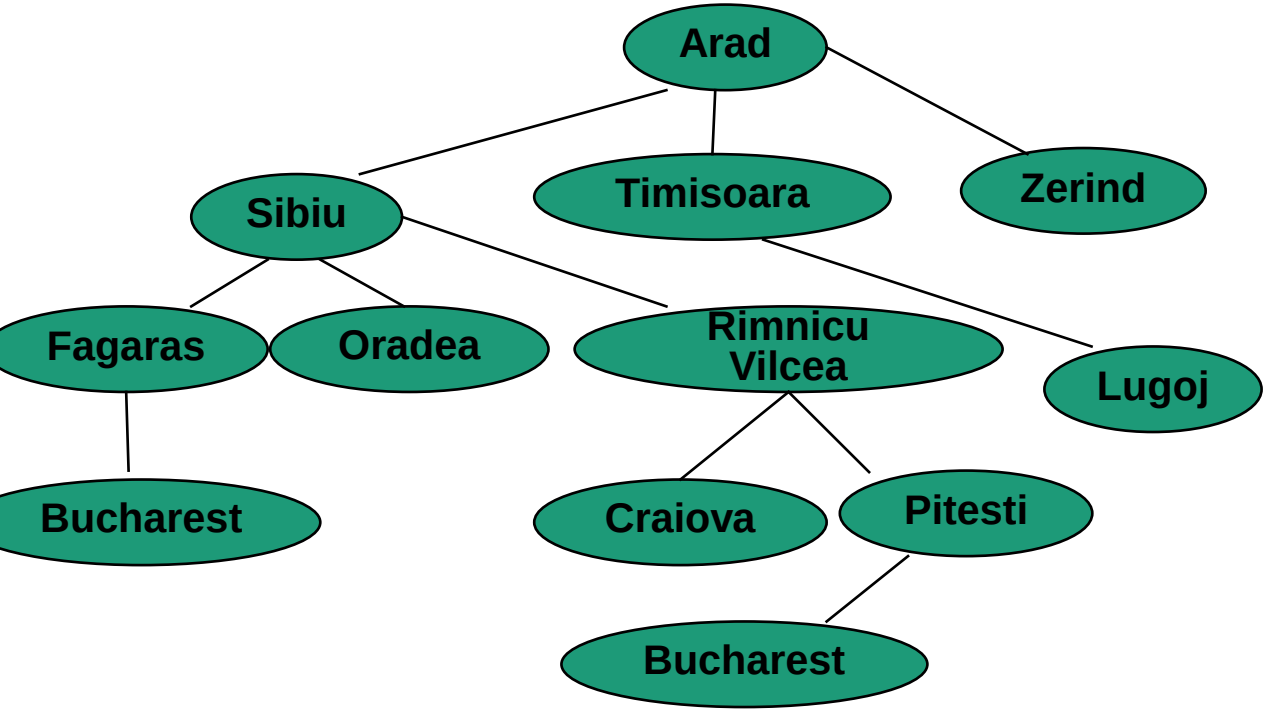
Algorithm

```
Let fringe be a list containing the initial state with depth = 0

loop do
    if fringe is empty then return failure
    Node            remove-first(fringe)
    if ~~Node~~ is goal state then
        return the corresponding solution
    else if depth of node < depth limit then
            expand all successor of Node and
            add the resulting(newly generated) nodes to the front of
            fringe with incremented depth
  end
```

Node will be expanded in a LIFO order.

DLS

Depth Limit : 2



Fringe	Closed
(A) (Expand d=0)	Nil
(Z T S)	(A)
(Z T RV O F)	(A S)
(Z T RV O)	(A S F)
(Z T RV O)	(A S F O)
(Z T)	(A S F O RV)
(Z)	(A S F O RV T)
()	(A S F O RV T Z)
	Goal Not Found

Path : NOT Found

DLS

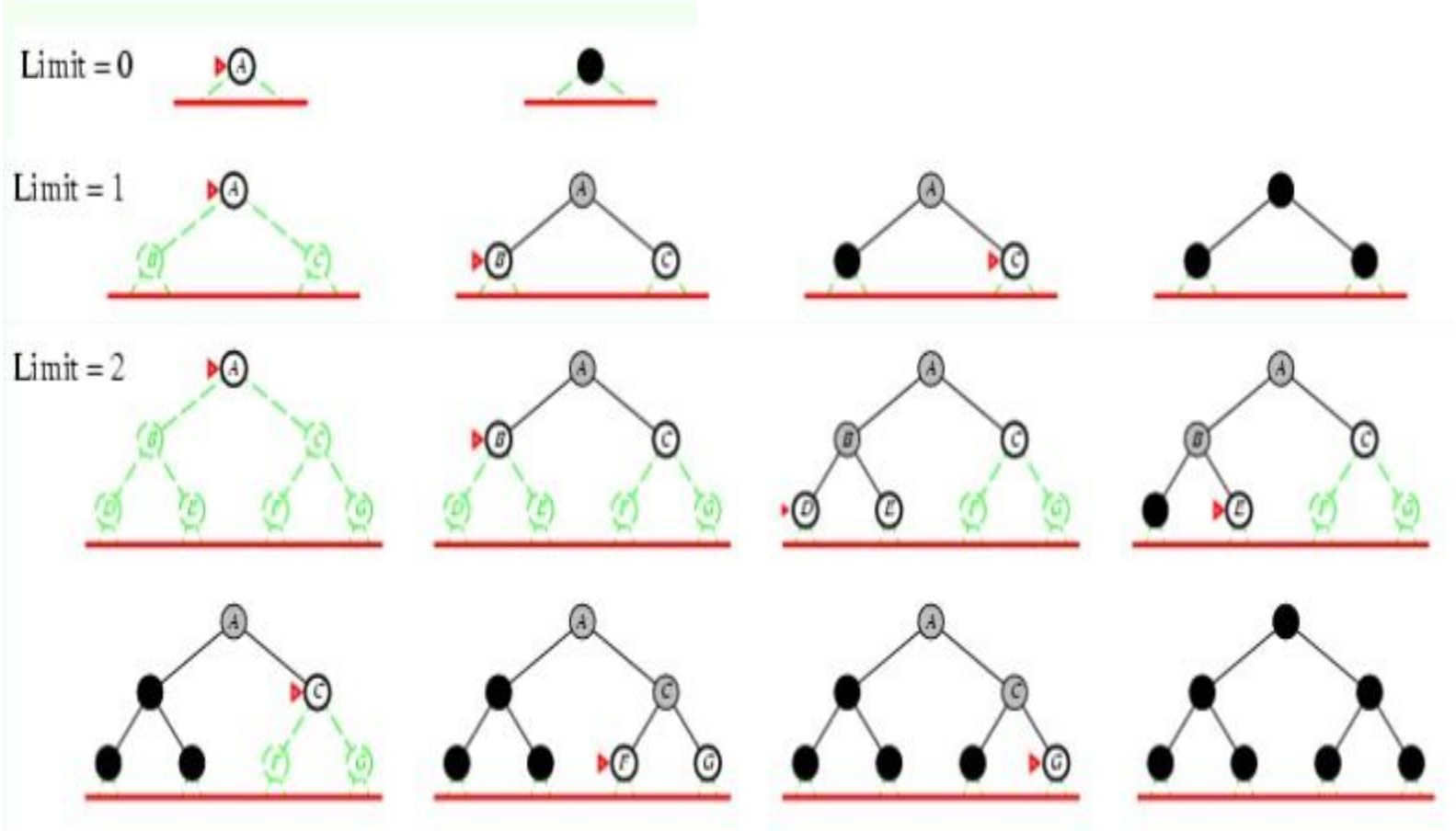
- **Advantages:**
  - Depth-limited search is Memory efficient.

- **Disadvantages:**
  - Depth-limited search also has a disadvantage of incompleteness.
  - It may not be optimal if the problem has more than one solution.

Iterative Deepening Search (IDS)

- Iterative deepening search is a depth-first tree search that determines the optimal depth limit by incrementally increasing the limit.
  - It begins with a depth limit of **0** and gradually increments it to **1**, **2**, and so forth, until a goal is found.
  - The process concludes when the depth limit equals **d**, the depth of the shallowest goal node.
  - IDS effectively combines the advantages of both depth-first search and breadth-first search.

Example - IDS



IDS

- **Advantages:**
  - It combines the benefits of BFS and DFS search algorithm in terms of fast search and memory efficiency.

- **Disadvantages:**
  - The main drawback of IDDFS is that it repeats all the work of the previous phase.

Problem

2		3
1	8	4
7	6	5

Initial State

1	2	3
8		4
7	6	5

Goal State

Goal formulation

Arrange 1-8 numbered tiles as per the sequence given in problem statement

Problem formulation

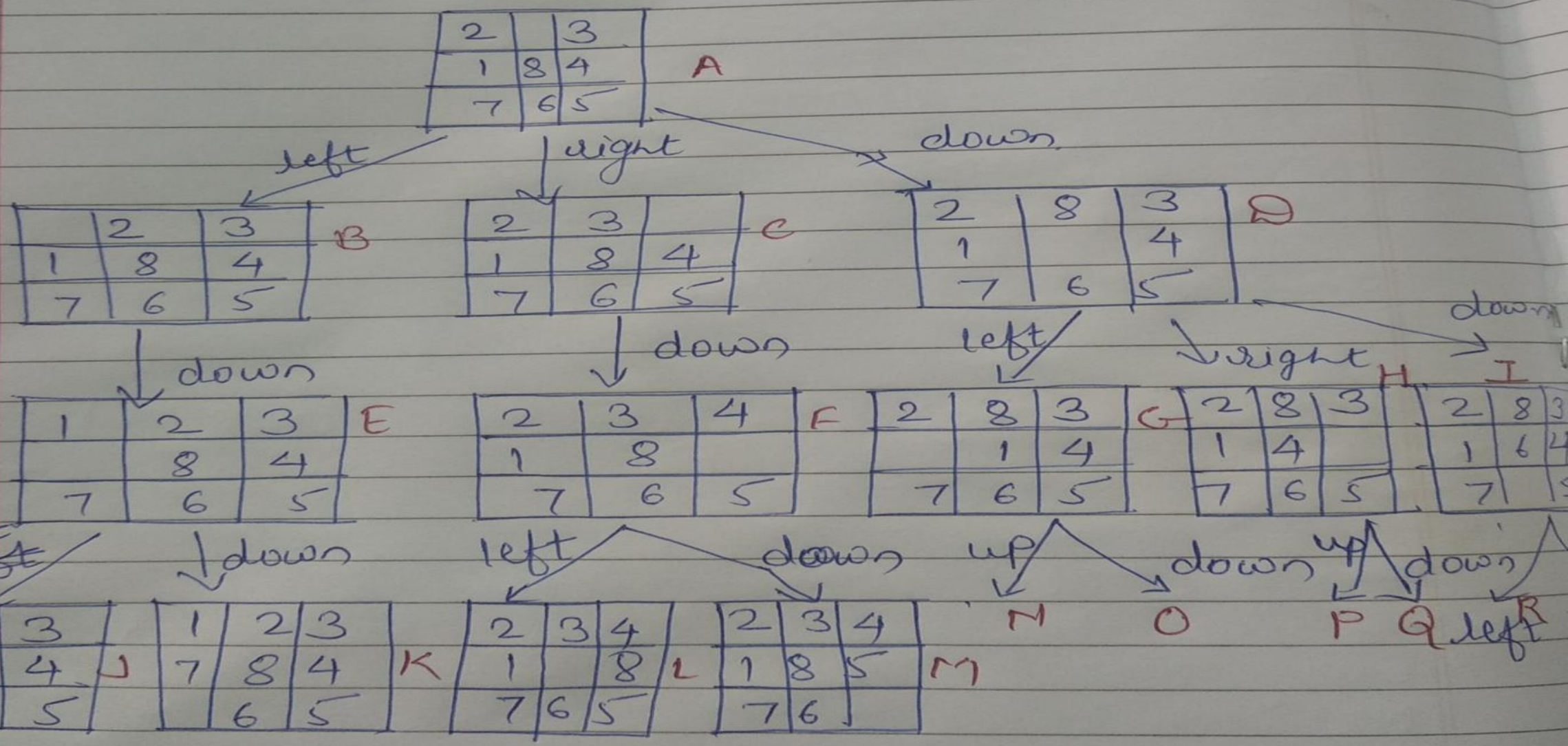
- State description- 3 X 3 grid of tiles numbered through 1 to 8 and a blank tile.
- Initial state – given combination of 3 X 3 grid
  - 8-puzzle(2,$,3,1,8,4,7,6,5)
- Actions- left(), right(), up(), down()
- Successor function (on goal state)
  - <left(), 8-puzzle(1,2,3,$,8,4,7,6,5)>
  - <right(), 8-puzzle(1,2,3,8,4,$,7,6,5)>
  - <up(), 8-puzzle(1,$,3,8,2,4,7,6,5)>
  - <down(), 8-puzzle(1,2,3,8,6,4,7,$,5)>

2		3
1	8	4
7	6	5

IS

1	2	3
8		4
7	6	5

GS



BFS, DFS, DLS, IDS

- BFS
- DFS
- DLS
- IDS

Puzzle

- Initial State

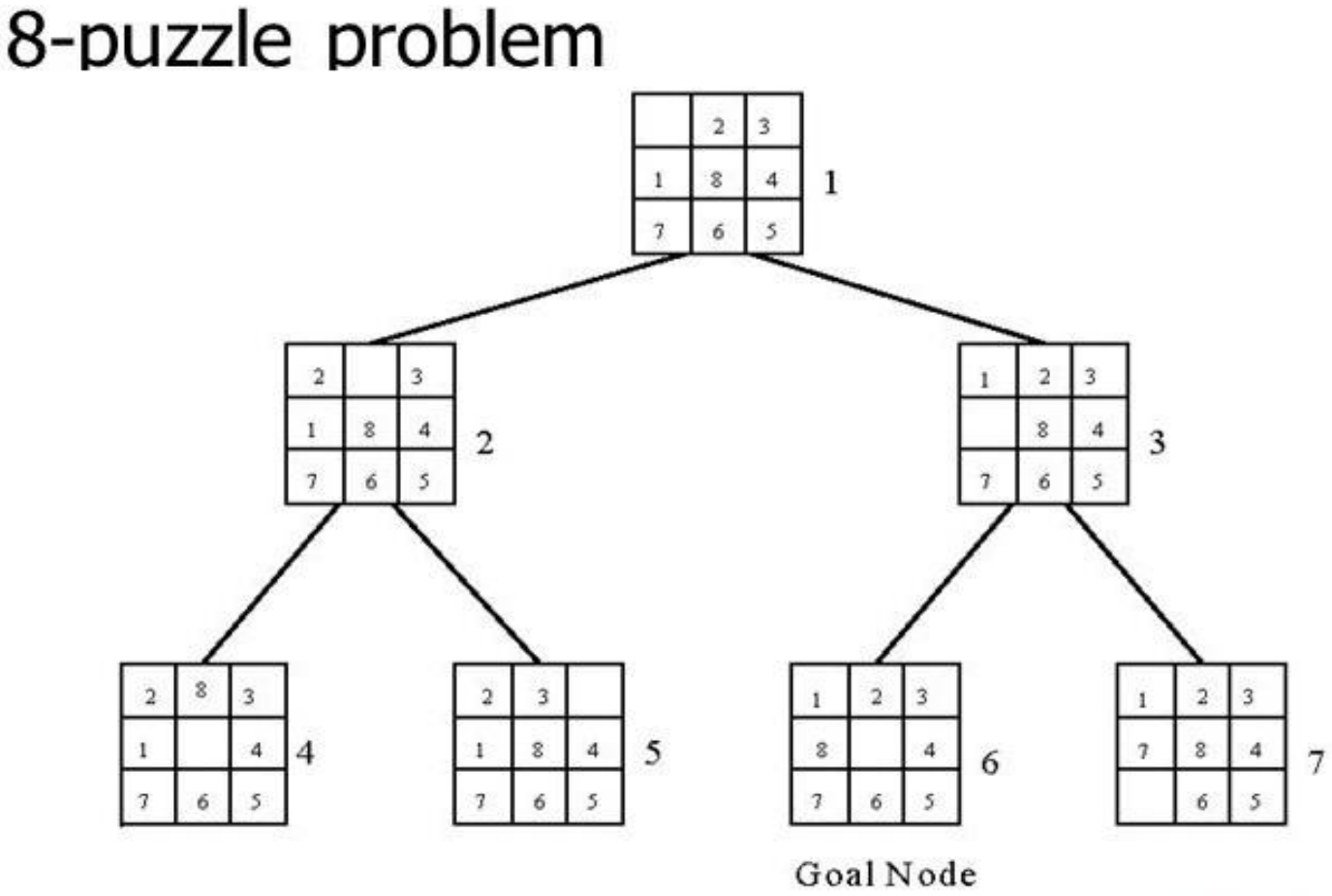
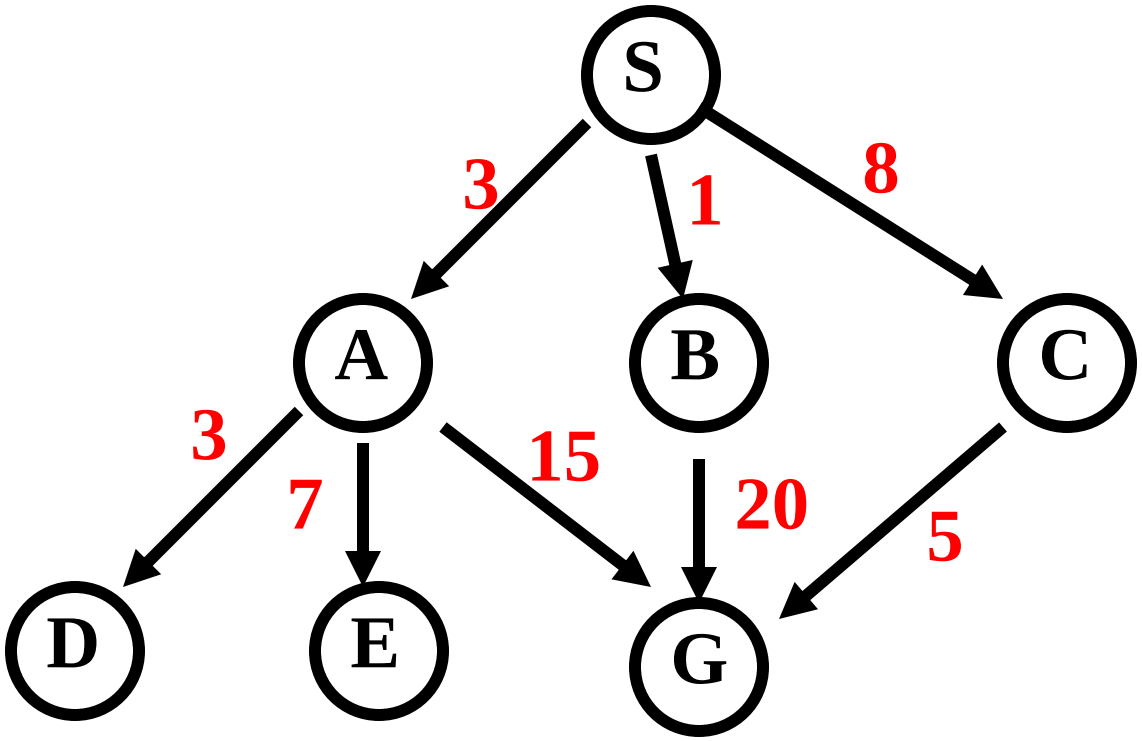
	2	3
1	8	4
7	6	5

- Goal State

1	2	3
8		4
7	6	5

State Space Tree

- Solve
  - BFS
  - DFS
  - DLS
  - IDS



Exercise